H-B-643c
June 2, 1966
SUPERSEDING
Fed. Spec. H-B-643b
April 30, 1959

FEDERAL SPECIFICATION

BRUSHES, ACID SWABBING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers acid swabbing brushes with various filler materials.

1.2 **Classification.**

1.2.1 *Types, classes, and sizes.* Acid swabbing brushes covered by this specification shall be of the following types, classes, and sizes as specified (see 6.2).

Type I—Round, twisted in wire.
- Class 1—Bristle or hair.
- Class 2—Synthetic filaments.

Type II—Flat, metal handle.
- Class 1—Bristle or hair.
- Class 2—Synthetic filaments.
- Size 1—3/8 inch (handle width, brush end).
- Size 2—7/16 inch (handle width, brush end).
- Size 3—9/16 inch (handle width, brush end).

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

H-S-951—Synthetic Filaments for Brushes; General Specification for.
R-N-91—Naphthalene, Technical.
PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-636—Box, Fiberboard.
PPP-B-665—Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676—Boxes, Setup.

Federal Standard:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Preproduction sample. When specified (see 6.2), before production is commenced, a sample consisting of one type I brush or thirty type II brushes covered by this specification, shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Standard product. The brushes delivered under this specification shall be the manufacturer's standard commercial product, except for any modification and additional requirements specified herein.

3.3 Materials. The materials used shall conform to the requirements specified hereinafter. Materials not definitely specified shall be of the quality normally used for the purpose in commercial practice.

3.3.1 *Filler material.* The filler material shall be as specified hereinafter. The filler material shall have no perceptible change in stiffness or evidence of artificial color when soaked in boiling water at 212° F. for not less than one hour when tested in accordance with 4.4.1. After removing the filler material from the water there shall be no visual evidence of water discoloration.

3.3.1.1 *Bristle or hair.* The bristle or hair shall be stiff hog bristle riflings or stiff horsetail hair thoroughly sterilized as specified in 3.3.1.1.1.

3.3.1.1.1 *Sterilization.* The horsetail hair and hog bristle shall be sterilized prior to manufacturing the brushes (see 6.4), after thorough washing and rinsing, in accordance with any one of the procedures specified in 3.3.1.1.1.1, 3.3.1.1.1.2 or 3.3.1.1.1.3.

3.3.1.1.1.1 *Boiling water.* Hog bristle and horsetail hair shall be soaked in boiling water (212°F.) for not less than 30 minutes.

3.3.1.1.1.2 *Dry heat.* Hog bristle and horsetail hair shall be placed in a chamber and a temperature above 284°F. shall be developed and maintained for not less than 3 hours.

3.3.1.1.1.3 *Autoclaving.* Horsetail hair and hog bristle shall be placed in a steam pressure chamber for a minimum of 45 minutes at not less than 15 pounds per square inch gage (p.s.i.g.).

3.3.1.2 *Synthetic filaments.* The synthetic filaments shall meet the requirements for type II, class "N" filaments of H-S-951, except for the requirements for modulus of elasticity. The diameter of the filament shall be 0.008 ± 0.001 inch.

3.3.1.3 *Metal.*

3.3.1.3.1 *Wire.* The wire in type I brushes shall be soft galvanized-steel wire not less than 0.052 inch in diameter.

3.3.1.3.2 *Handle.* The handle of the type II brushes shall be fabricated from tin and shall have a 1/2 pound electrolytic tinplate in accordance with the manufacturer's commercial practice.

3.4 Construction. The type I brush shall have a rounded brush head fabricated from filler material specified in 3.3.1, and two strands of wire twisted together forming the handle. The handle shall have an integral eye or loop. The type II brush shall be constructed of a tin-plated tubular handle specified in 3.4.2. The filler material shall extend from one end of the tubular handle. The weight of the filler material for the type II brushes shall be as required by table II when tested as specified in 4.4.3. The figure incorporated in this specification is for illustration and convenience of identification only and is not intended to preclude the purchase of brushes otherwise in accordance with the requirements of this specification.

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3.4.1 *Type I.* The type I brushes shall conform to the physical requirements in table I and shall be similar to figure 1. The filler material for the class 1 brush shall be bristle or hair specified in 3.3.1.1 and the filler material for the class 2 brush shall be synthetic filament specified in 3.3.1.2. The filler material shall be secured by a two stranded twisted wire core and shall project perpendicular to the wire core. The wire strands specified in 3.3.1.3.1 shall be firmly twisted to not less than 5 nor more than 8 turns per inch of length to prevent the filaments from pulling out of the brush when tested in accordance with 4.4.2. All exposed parts of the wire shall be rounded and smooth. At the handle, there shall be a 3/4 inch minimum diameter integral eye or loop formed by the handle wire and shall not show any breaks, cracks, or untwist when tested as specified in 4.4.4.

TABLE I. *Physical requirements for type I brushes*

Length, overall, minimum inches ...	8 1/4
Brush parts:	
Diameter, minimum, inch	7/16
Length, minimum, inches	2-7/8

3.4.2 *Type II.* The type II brushes shall conform to the physical requirements in table II and shall be similar to figure 1. The filler material for the class 1 brush shall be bristle or hair specified in 3.3.1.1 and the filler material for the class 2 brush shall be synthetic filament specified in 3.3.1.2 and held in a tin-plated tube handle specified in 3.3.1.3.2 by a staple or set in rubber and secured by clinching or stamping the tube at the brush end. In addition, the seam of the tube for class 2 brushes shall be spot welded or soldered at the brush end where the synthetic filaments are inserted. When tested as specified in 4.4.2, it shall not be possible to remove the filler material from the handle part by pulling with a force of 8 pounds.

TABLE II. *Physical requirements for type II brushes*

	Size 1	Size 2	Size 3
Length, overall, minimum, inches:	5 3/4	5 3/4	5 3/4
Handle, length, minimum, inches:	5	5	5
Handle, width, hand end, minimum, inch:	1/4	5/16	3/8
Handle, width, brush end, minimum, inch:	3/8	7/16	9/16
Length, filament, bristle or hair clear of handle, minimum, inch:			
Class 1	23/32	25/32	27/32
Class 2	3/4	3/4	3/4
[1] Bristle per 100 gross, minimum pounds:	21	33	46
[2] Filament per 100 gross, minimum pounds:	18	28	38
Metal thickness:	0.010 ± .001	0.011 ± .001	0.012 ± .001

[1] Weight of bristle or hair per 15 brushes: Size 1, 153 grains; size 2, 240 grains; size 3, 335 grains, when tested in accordance with 4.4.3.

[2] Weight of filament per 15 brushes: Size 1, 131 grains; size 2, 204 grains; size 3, 277 grains, when tested in accordance with 4.4.3

3.5 Workmanship. The finished brush shall conform to the quality and grade of product established by this specification. There shall be no exposed sharp wire, edges, or splinters.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Preproduction sample inspection. When a preproduction sample is required, it shall be examined for defects in table III and dimensions specified herein and shall be tested as specified in 4.4.3 and 4.4.4.

4.3 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.3.1 *Component and material inspection.* In accordance with 4.1, the components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 *End item inspection.* A lot shall consist of all the brushes of the same type, class, and size offered for inspection at one time. The sample unit shall be one completely fabricated brush unless otherwise specified.

4.3.2.1 *Visual examination.* Examination shall be made of the end item for the defects listed in table III. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects, and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE III. *Classification of defects*

Examine	Defect	Classification	
		Major	Minor
Workmanship (applicable to all components and assemblies) (Type I)	Sharp wire edge or splinter	X	
	Wire strands not firmly and uniformly twisted	X	
	Bristles or filaments, as applicable, can be pulled out of handle by hand	X	
	No integral eye or loop		X
	Any exposed area of wire not rounded and smooth		X
(Type II)	Clinching or stamping of tube at brush end not uniformly and securely accomplished	X	
	Brush not stapled or set in rubber in handle	X	
	Seam of handle not spot welded or soldered at brush end when brush consists of synthetic filaments	X	
	Any exposed sharp or rough surfaces or exposed splinter ...	X	
Handle (Type I)	Not two stranded twisted-wire core ..	X	
	Not galvanized		X
	Less than 5 turns per inch, more than 8 turns per inch		X
(Type II)	Not tubular	X	
	Not tinplated		X
Bristles (Class 1)	Not stiff hog bristles or horsetail hair ...	X	
(Class 2)	Not synthetic filaments	X	

4.3.2.2 *Dimensional examination.* Inspection shall be made of the end item for compliance with dimensions specified. Any noncompliance from specified requirements shall constitute a defect. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

4.3.2.3 *Testing of the end item.* The end item shall be tested for the applicable characteristics as indicated in table IV. The sample unit for the test specified in 4.4.3 shall be 15 brushes of the same type, class, and size offered for inspection at one time.

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TABLE IV. *Instructions for testing*

Characteristic	Specification reference		Requirements applicable to		Number determinations per unit	Results reported as		In-spect. level	AQL
	Require-ment	Test method	Indi-vidual brush	Lot aver.		Pass or fail	Numer-ically to nearest		
Soak test	3.3.1	4.4.1	X	—	1	X	—	S-2	4.0
Set retention test	3.4	4.4.2	X	—	1	X	—	S-2	4.0
Weight of bristle or hair	Table II	4.4.3	X	—	2[1]	—	One grain	N/A	N/A
Weight of filament	Table II	4.4.3	X	—	2[1]	—	One grain	N/A	N/A
Bend test	3.4	4.4.4	X	—	1	X	—	S-2	4.0

[1] The sample size shall be two units, 15 brushes per unit. The composite weight of bristle, hair or filament, as applicable for 15 brushes (each sample unit), shall be obtained. The average of the composite weight shall be reported to the nearest grain for compliance to table II.

4.3.2.4 *Examination of preparation for delivery.* An examination shall be made to determine that packaging, packing, and marking requirements contained in section 5 are met. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery and selected at random just prior to the closing operations. The lot shall be the number of containers offered for delivery at one time. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units. Defects shall be as specified in table V. Examination for closure and strapping shall be in accordance with applicable container specification.

TABLE V. *Examination of preparation for delivery*

Examine	Defect
Markings (exterior and interior containers)	Missing, incorrect, illegible, or improper size, location, sequence, or method of application.
Material	Component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling, bulging, or distortion of containers.
Contents	Missing or inadequate quantity of moth repellent naphthalene balls or flakes for class 1 brushes.
	Intermediate packaging containers are more or less than required.

4.4 Tests.

4.4.1 *Soak test.* Soak the filler material in boiling water at 212°F. for not less than one hour. Remove the filler material from the water and place it face up in a holding device so that it is held firmly in a manner and position similar to a finished brush. A finished brush shall be attached in a face up position alongside the filler material. The filler material in the holding device and the brush shall be brushed with the palm of the hand for comparison. The filler material and the water used for the test shall be examined for compliance with 3.3.1.

4.4.2 *Set retention.* Grasp the filler material tuft with a suitable holding device and applying a force, by using a spring scale or weight of eight pounds in the direction tending to pull the tuft loose and examine the filler material to determine compliance with 3.4.1 and 3.4.2.

4.4.3 *Weight of filler material.* Remove the bristle or horsehair, class 1, filler material from the brush and condition for not less than 16 hours in air at a relative humidity of 50 ± 4 percent and at a temperature of 23° ± 1.1°C. Weigh the filler material in the same atmosphere for compliance with table II. Remove the filament, class 2, filler material from the brush and weigh the filler material in the same atmosphere for compliance with table II.

4.4.4 *Bend test for handle (type I only).* Bend the handle to an angle of 180° over a

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mandrel with a diameter equal to that of the handle. Examine the handle to determine compliance with 3.4.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.* One hundred and forty-four brushes of one type, class, and size only, shall be packaged in a box conforming to PPP-B-566, PPP-B-665, or PPP-B-676. Each box with class 1 bristle or hair shall contain an adequate quantity of naphthalene balls or flakes conforming to R-N-91. Each box shall be securely closed in accordance with the appendix of PPP-B-566 or PPP-B-676.

5.1.2 *Level C.* Brushes shall be packaged to afford adequate protection against deterioration and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice, when it meets these requirements.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.* Twenty unit packages of brushes (2,880 brushes) of one type, class, and size only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V3c or V3s of PPP-B-636. Closure, waterproofing by means of tape, and reinforcing with flat steel strapping or tape banding shall be in accordance with the appendix of the container specification.

5.2.2 *Level B.* Unless otherwise specified, twenty unit packages of brushes (2,880 brushes) of one type, class, and size only, packaged as specified in 5.1, shall be packed in a snug-fitting box conforming to PPP-B-636, class domestic. When specified (see 6.2), the container shall conform to V3c or V3s of PPP-B-636. Closure shall be in accordance with the appendix of the container specification.

5.2.3 *Level C.* Brushes, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 Standard pack, civil agencies. The standard pack for civil agency procurements shall be as specified in 5.1.1 and 5.2.2.

5.4 Marking.

5.4.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 *Military requirements.* In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The brushes covered by this specification are primarily intended for swabbing solderings, for seam sealing, and for applying oil to metal markings. Class 2 synthetic filament brushes are not recommended for use with strong acids for long periods of time.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, and size brush required (see 1.2).
(*c*) When preproduction sample is required (see 3.1).
(*d*) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(*e*) If weather-resistant fiberboard is required for level B pack (see 5.2.2).

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6.4 Certificate of compliance. At the option of the contracting officer, the horsehair or bristle may be accepted on the basis of a supplier's certificate of compliance for the sterilization requirements specified in 3.3.1.1.1.

6.5 Standard pack for civil agencies. The standard pack requirements in 5.3 are intended for use in procurements of stores stock replenishments. Procuring officers should use the standard pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot, or intermediate storage point for temporary storage, subsequent issue, and shipment to eventual user.

6.6 Level B packaging. When level B packaging is specified for civil agency procurement, the requirements of paragraph 5.1.1, level A packaging, shall apply.

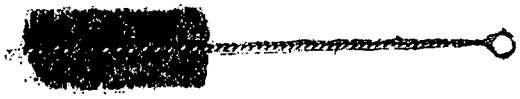
CUSTODIANS:
Army—GL
Navy—SA
Air Force—69

Review activities:
Army—GL
Navy—SA
Air Force—69

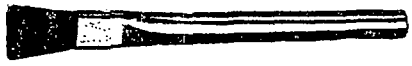
User activities:
Army—CE, MO, SM
Navy—CG, MC

Preparing activity:
Army—GL

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Type I. Rounded, twisted in wire.



Type II. Flat, metal handle.

FIGURE 1. Acid-swabbing brushes

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 5 cents each.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION
Brushes, Acid Swabbing - H-B-643c

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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